SPEECH SIGNAL ANALYSIS

Laboratory report

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2 Bandwidth of the speech signal

Open the file and visualize both the waveform and the spectrogram of the signal. Answer the following questions:

1. What is the sampling frequency? (be sure to visualize the full spectral range). How do you know?

The audio that I used is: **AGC0003_neu.wav.**

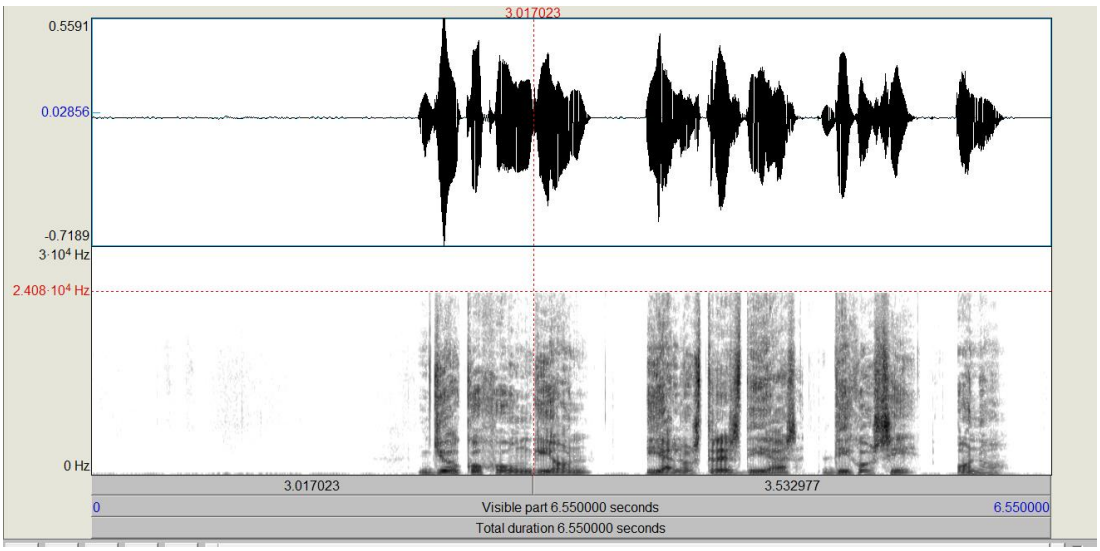
The sampling frequency can be checked in two ways. The first one by clicking into the "info" button in Praat objects. We can observe in the following screenshot that it is 48000 Hz.

Figure 1

```
Number of channels: 1 (mono)
Time domain:
   Start time: 0 seconds
   End time: 6.55 seconds
   Total duration: 6.55 seconds
Time sampling:
   Number of samples: 314400
   Sampling period: 2.0833333333333333e-05 seconds
   Sampling frequency: 48000 Hz
   First sample centred at: 1.0416666666666666e-05 seconds
```

The second way to check it is by visualizing the spectrogram itself. According to the Nyquist theorem, the maximum frequency that a sampled signal can contain is fs/2. By configuring the spectrogram with a view range of 0-30000 Hz, a clear limit is observed: the signal contains information only up to 24000 Hz, while the region between 24000 and 30000 Hz is completely empty. This confirms that the Nyquist frequency is 24000 Hz, and therefore the sampling frequency is 24000 x 2 = 48000 Hz

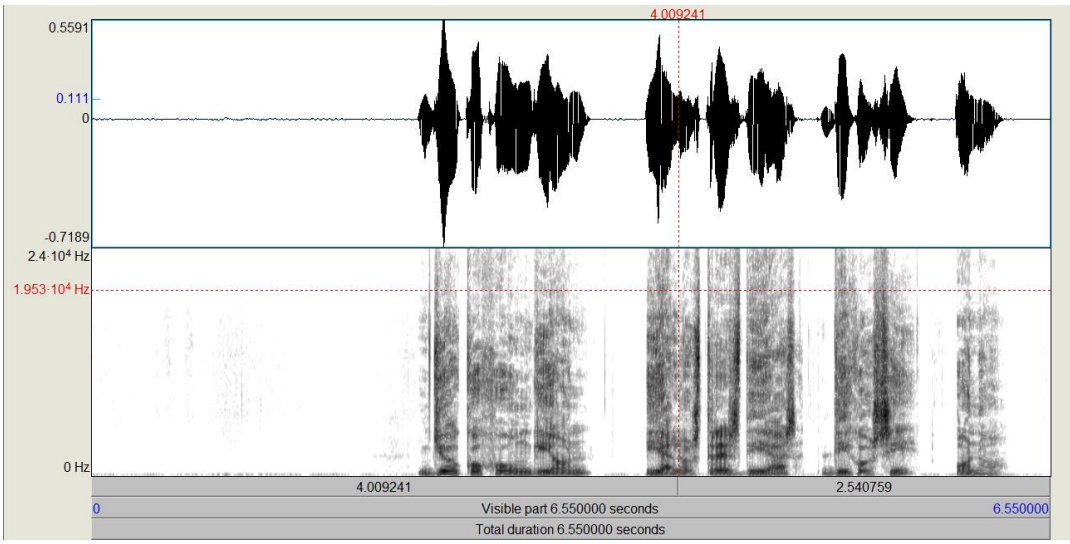
Figure 2



2. Observe the spectrogram (pay attention to seeing the full range of frequencies). Up to what frequency would you say that there is 'information' in the spectrogram? Let's call that frequency fmax.

Observing the spectrogram with a view range set to the full spectral range, two regions can be clearly distinguished: Below approximately 19 kHz, the spectrogram shows a clear contrast between speech segments and silence regions. Speech segments appear as dark, densely structured columns, while silent regions appear lighter. Within the speech columns, short white gaps are also visible, corresponding to natural brief silences in speech. The presence of these gaps confirms that meaningful information is still being captured in this frequency range. The contrast is strongest in the low and mid frequency regions (below 10kHz) and progressively weaker between 10 kHz and 19 kHz. Above approximately 19 kHz, the spectrogram becomes visually uniform: the short white gaps disappear and speech and silence regions look almost identical. This indicates that the voice no longer contributes significant energy above this frequency. Therefore, fmax is approximately 19kHz (see Figure 3).

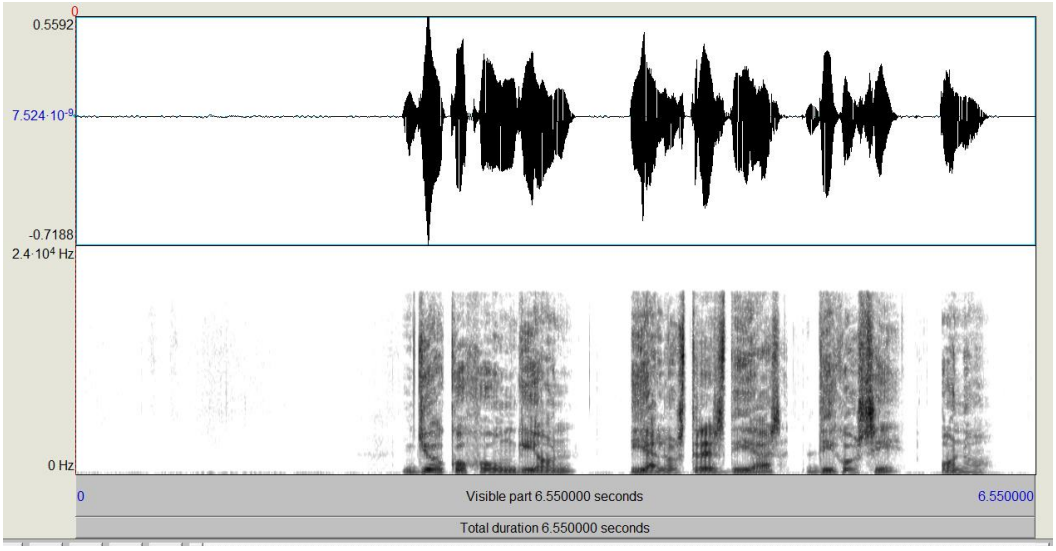
Figure 3



3. Filter your signal with a low-pass filter with a cut-off frequency equal to fmax. Listen to your filtered signal. Can you notice the difference?

After filtering the signal, the perceived difference when comparing the filtered signal to the original is almost imperceptible. Both recordings sound essentially the same, perhaps with the filtered signal sounding very slightly cleaner, but almost unnoticeable. The original recording already had a very low noise level, which maybe is why the difference is barely noticeable.

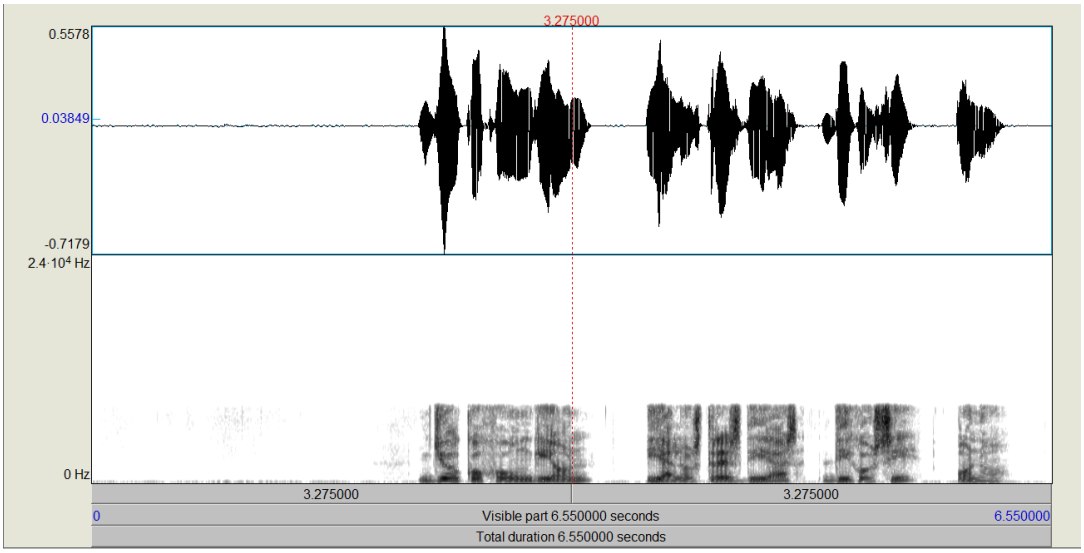
Figure 4: Spectrogram on the filtered signal, low pass filter.



4. Further filter the signal (you can further filter the already filtered signal or start from the original signal) now using a cut-off frequency of 8000 Hz. Can you notice the differences with the previous signals? This frequency value of 8000 Hz is a very common value used as Bandwidth in speech processing tasks (such as speech synthesis, speech recognition etc.).

After filtering the signal again, the perceived difference is now clearly noticeable, unlike in the previous case. The speech remains fully intelligible (all words can be understood without any difficulty) but the overall sound quality in noticeably reduced: the voice sounds less crisp and less "bright" compared to the original and to the 19 kHz version.

Figure 5: Spectrogram on the filtered signal, 8000 Hz



5. What is the minimum sampling frequency to be used with a signal with that bandwidth of 8000 Hz?

According to the Nyquist theorem, the sampling frequency must be at least twice the maximum frequency present in the signal. For a signal with a bandwidth of 8000 Hz, the minimum sampling frequency is therefore 16000 Hz, as a result of 2 x 8000.

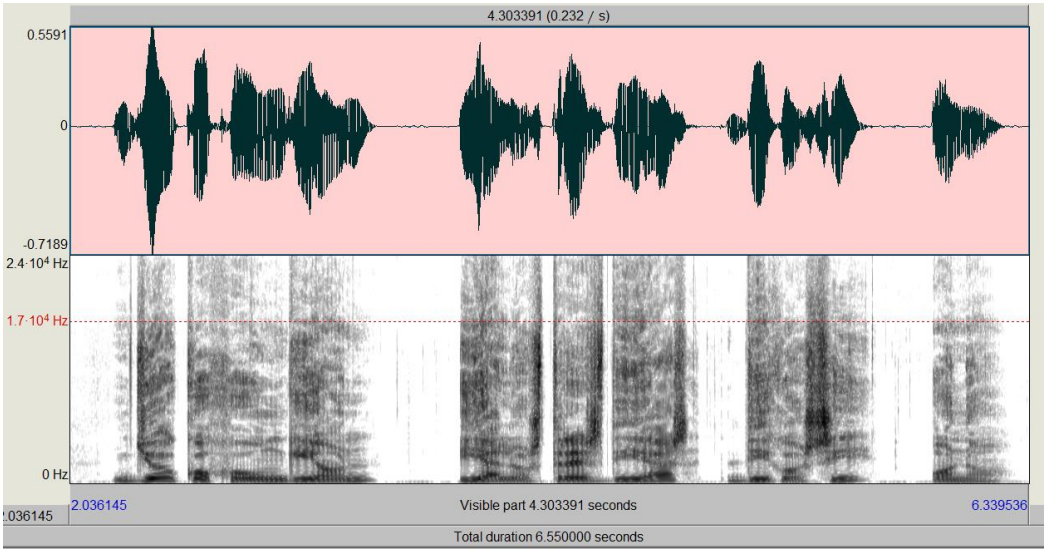
For the following sections, you can continue using the previous high quality signal (the original) or you can use one of your own. In this last case, consider that the signal should offer good quality up to 8000 Hz.

6. What are the sounds (phones) that occupy the maximum bandwidth? What kind or kinds of sounds reach the highest frequencies in your sentence?

I'm using the same audio file for the rest of the exercise (**AGC0003_neu.wav)**

The sounds that occupy the maximum bandwidth in the sentence are the unvoiced fricatives, mainly the /s/ sounds, as they distribute their energy across a very wide range. In the spectrogram (Figure 6), they appear as narrow, noisy columns clearly higher than the rest of the phones. The kinds of sounds that reach the highest frequencies are therefore the unvoiced fricatives as well, since they can extend beyond 15 kHz, while voiced sounds, concentrate most of their energy below, 4 kHz where the formants are located.

Figure 6: Zoom in the spectrogram

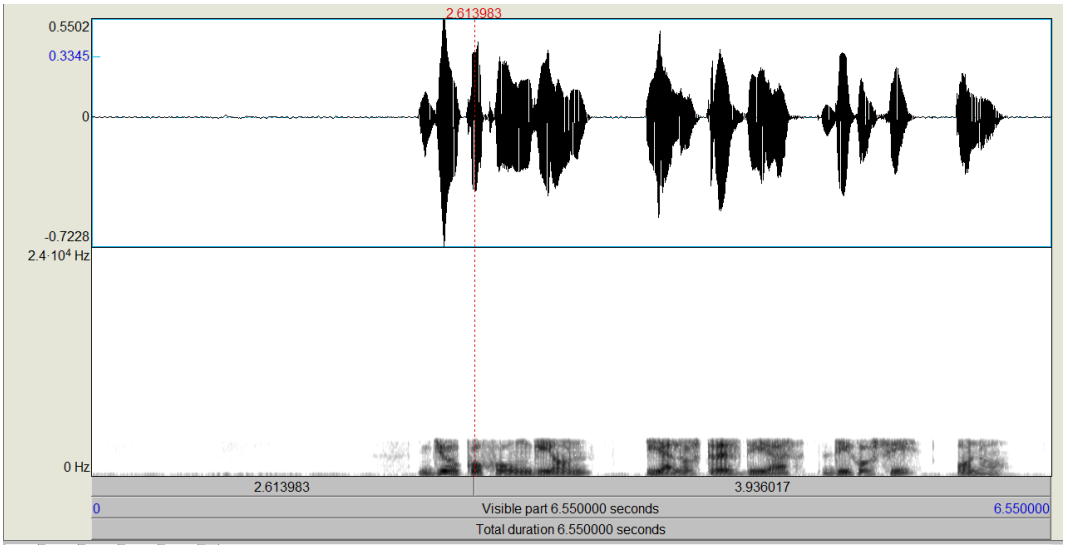


Use a low-pass filter to filter your signal to a bandwidth of 3600Hz. Save this filtered signal for the next steps. Listen to the differences between the original sentence and the one filtered at 3600 Hz.

7. Which sounds have been affected the most?

After filtering the signal, the most affected sounds are the /s/ sounds. They can still be perceived but have clearly lost their characteristic sharpness. In contrast, vowels and voiced consonants remain perfectly recognizable.

Figure 7: Low pass filter 3600Hz



Low-pass filter the signal using a decreasing cutoff frequency value, i.e., progressively reducing the bandwidth of the signal (for example, starting from the previous signal filtered at 3600 kHz using steps of 400 Hz). For each filtering step, listen carefully and observe how some sounds cannot be distinguished anymore due to the filtering.

8. Approximately, which cut-off frequency makes the sentence unintelligible? It is better to check this fact with someone who does not know the contents of the sentence. Start with the lowest cut-off frequency (less intelligibility) until your partner can understand the content of the sentence.

The signal was filtered progressively with decreasing cut off frequencies (2400 Hz, 1200 Hz, 1000 Hz, 800 Hz, 600 Hz). The filtered signals were played in increasing order of cut-off frequency. At 600 and 800 Hz, the listener could not understand anything. At 1000 Hz, the sentence was still unintelligible for the listener. At 1200 Hz, the listener could partially recover the content with some effort. At 2400 Hz, the sentence was clearly intelligible although noticeably muffled.

Therefore, the approximate cut of frequency at which the sentence becomes unintelligible is around 1000 Hz.

9. Take the signal filtered at 3600 Hz and now use a 'high pass filter' with cut-off frequency 300 Hz to filter that signal. Can you still identify the message? Can you also identify the speaker? Comment on this process.

Yes, the message can still be clearly identified, although with less clarity than in the original. The speaker is also identified as the same female voice. The resulting signal sounds very similar to a telephone or a walkie-talkie recording.

This works because the 300·3600 Hz range retains F0 and the main vowel formants (F1, F2, F3), enough for both intelligibility and speaker recognition.

3 Waveform, spectrogram and spectrum of voiced and unvoiced sounds

To do this exercise you can use your own recording or a provided signal of your choice. In either case, be sure that the sentence is phonetically rich (i.e., it contains a variety of sounds). You can find some examples in section 6.4.

The goal of this exercise is to clarify the differences for voiced and unvoiced sounds, while visualizing three different representations: the waveform, the spectrogram and the spectrum of a segment. Pay attention to which representation is being asked.

Locate voiced and unvoiced segments in your speech signal. Use zoom to see the signal waveform properly, showing approximately one syllable. Note the differences in the waveform and in the spectrogram of both types of sounds:

Note: own recording of "The birch canoe slid on the smooth planks" (Harvard sentence set), recorded at 44,100 Hz in Praat.

1. What differences can you observe in time (waveform) and in frequency (spectrogram)? Show at least one example each (voiced/unvoiced waveform and spectrogram).

In the time domain, voiced sounds show a clearly periodic waveform. Repeating cycles of the same shape are visible, produced by regular vibration of the vocal folds. Unvoiced sounds have an irregular, noise-like waveform with no repeating pattern and generally lower amplitude.

In the spectrogram, voiced sounds produce horizontal striations across the frequency range. These are the harmonics of F0, each appearing as a dark horizontal band. Formant bands (F1, F2, F3) are also visible as regions of concentrated energy. Unvoiced sounds show diffuse, grainy energy with no horizontal structure, typically concentrated at higher frequencies. The pitch track (blue line) is present over voiced regions and absent over unvoiced ones.

Figure 1 shows the narrowband spectrogram of the full sentence. Voiced regions such as the vowels in "canoe" and "smooth" are clearly visible as areas of horizontal striations in the lower frequency range. Unvoiced consonants such as the /s/ in "slid" and the /pl/ cluster in "planks" appear as darker vertical bands of diffuse high-frequency noise where the harmonic striations disappear and the pitch track is absent. Figure 2 shows a zoomed view of the voiced vowel /u:/ in "smooth", where the periodic waveform structure and harmonic striations are clearly visible.

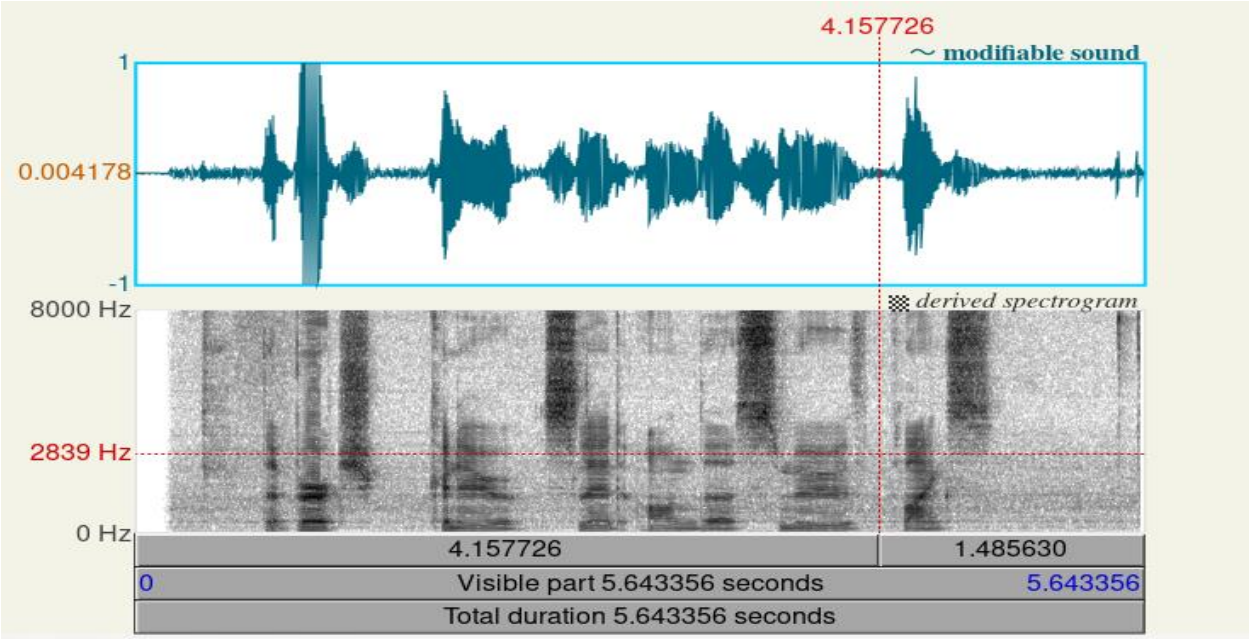


Figure 1. Narrowband spectrogram (window = 0.02 s) of the full sentence. Horizontal striations indicate voiced regions; diffuse vertical bands indicate unvoiced consonants.

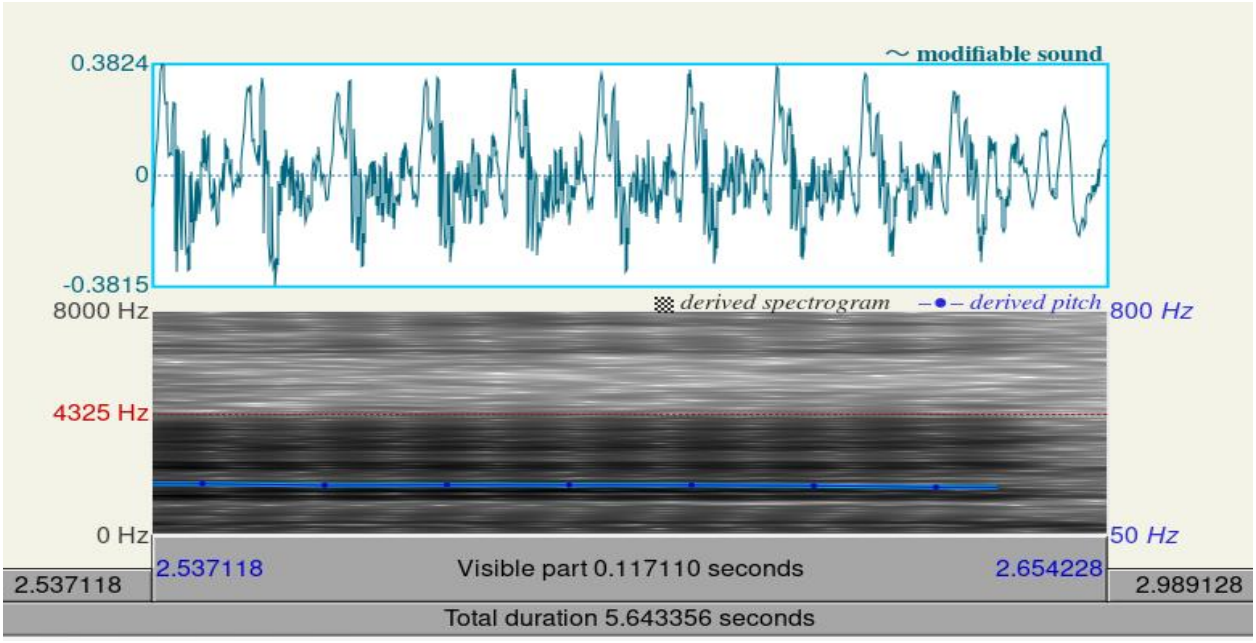


Figure 2. Zoomed view of the voiced vowel /u:/ in “smooth”. The waveform (top) shows clear periodic cycles. The spectrogram (bottom) shows horizontal harmonic striations and the blue pitch track confirming voicing.

Compare also the spectrum of a voiced sound with the one of an unvoiced sound. To do this, plot the Spectrum of a characteristic segment (voiced/unvoiced) (see section 6.2 if you need help to plot the spectrum). To obtain the spectrum use the parameters for a narrowband spectrogram.

2. What are the differences between the spectrum of voiced and unvoiced sounds? Explain them with detail and show one example.

The spectrum of a voiced sound shows regularly spaced peaks at multiples of F0, forming a harmonic series (Figure 3). The peaks here are about 95 Hz apart, which is F0. The formants are not the individual harmonic peaks — they are the broad bumps in the spectral envelope that sits over the harmonics. To read F1 and F2 you look at which groups of harmonics are being boosted. For /u:/ in "smooth", the first boosted cluster is around 285–380 Hz, so F1 is roughly 300–350 Hz, and the next cluster is around 900–1000 Hz, giving F2 at about 950 Hz. Both a low F1 and a low F2 are expected for /u:/, so these values make sense. Reading formants this way is not very precise since the harmonics are quite widely spaced at this F0 — the LPC method used in Section 4 gives cleaner estimates. Energy drops off at higher frequencies because of the glottal source rolloff.

The spectrum of an unvoiced sound shows broadband, noise-like energy with no harmonic peaks (Figure 4). For the fricative /s/ in “slid”, energy is concentrated mainly in the higher frequency range and there is no clear structure in the envelope. The absence of a harmonic series is the key difference from a voiced sound.

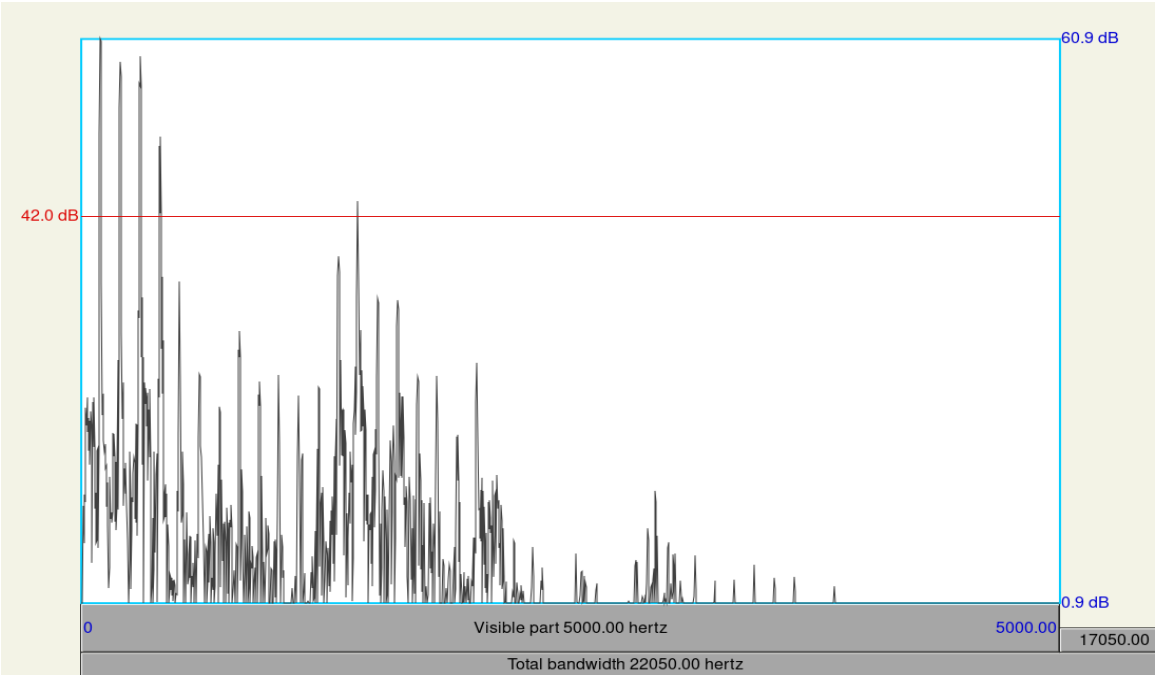


Figure 3. Spectral slice of the voiced vowel /u:/ in “smooth” (narrowband, 0--5000 Hz). Regularly spaced harmonic peaks are visible, spaced ~95 Hz apart (F0). The spectral envelope has its first peak (F1) around 300--350 Hz and its second peak (F2) around 950 Hz. Energy rolls off at higher frequencies.

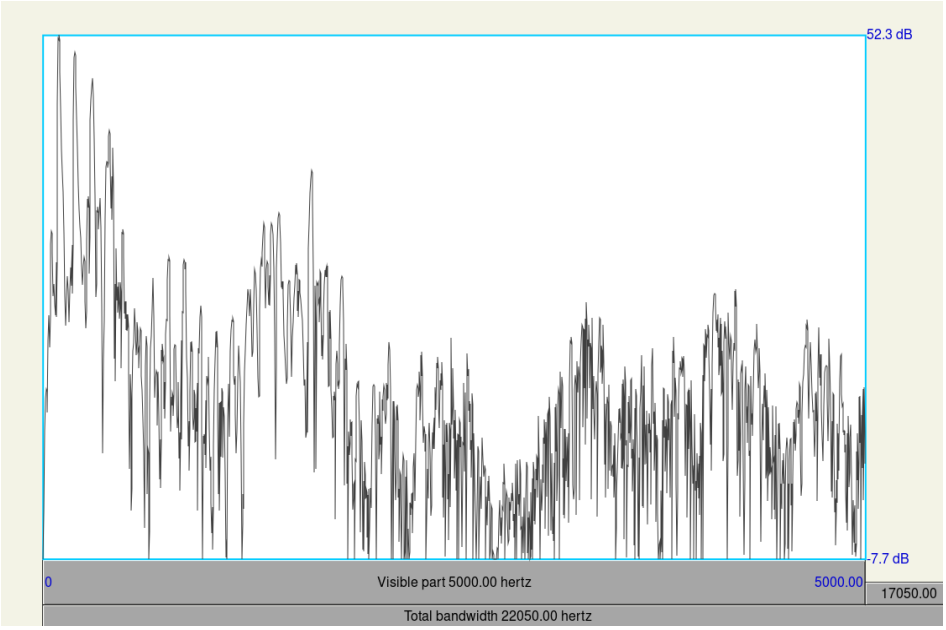


Figure 4. Spectral slice of the unvoiced fricative /s/ in "slid". The spectrum shows no harmonic peaks and no formant structure, consistent with aperiodic, noise-like frication typical of sibilants. Note that the view is limited to 0–5000 Hz; the characteristic high-frequency energy concentration of /s/ extends beyond this window.

3. Why should we use the parameters of a narrowband spectrogram to see voicing in the spectrum and not those of a broadband spectrogram?

To see individual harmonics in the spectrum you need enough frequency resolution to separate them. A 20 ms window gives a resolution of about 1/0.02 = 50 Hz, which is fine since this speaker's harmonics are spaced roughly 93 Hz apart.

With a 3 ms window the resolution drops to about 1/0.003 = 333 Hz. That is too coarse to separate harmonics only 93 Hz apart — they blur together and the harmonic structure disappears, so you cannot confirm voicing from the spectrum. The broadband setting is more useful for other things, like seeing individual glottal pulses as vertical striations or tracking fast formant movements, but it is the wrong choice here.

Figures 5 and 6 show the same recording under broadband and narrowband settings respectively. In Figure 5, vertical striations are visible but no horizontal harmonic structure. In Figure 6, clear horizontal harmonic bands are resolved across the voiced regions.

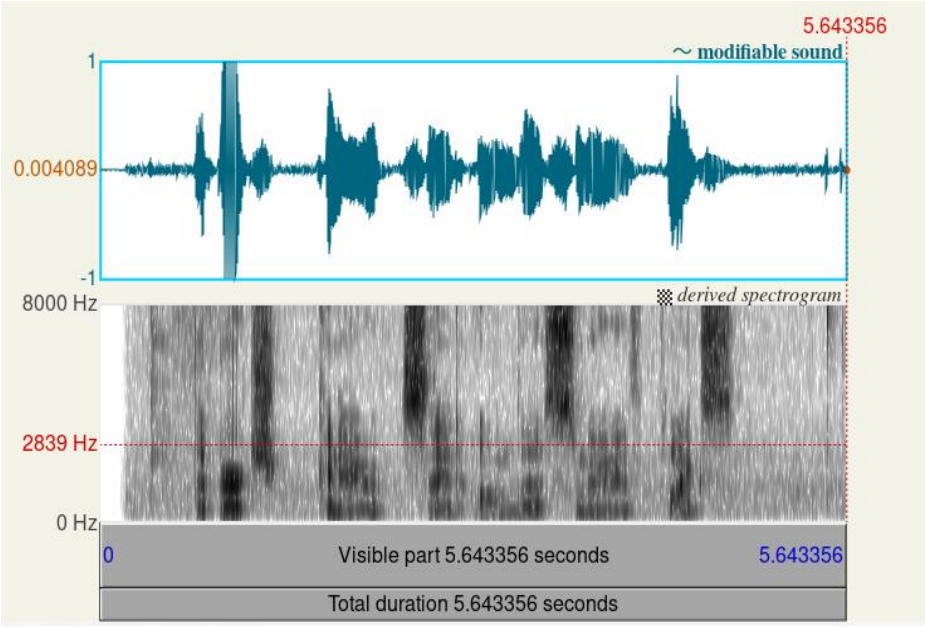


Figure 5. Broadband spectrogram (window = 0.003 s). Vertical striations are visible but individual harmonics cannot be resolved due to poor frequency resolution.

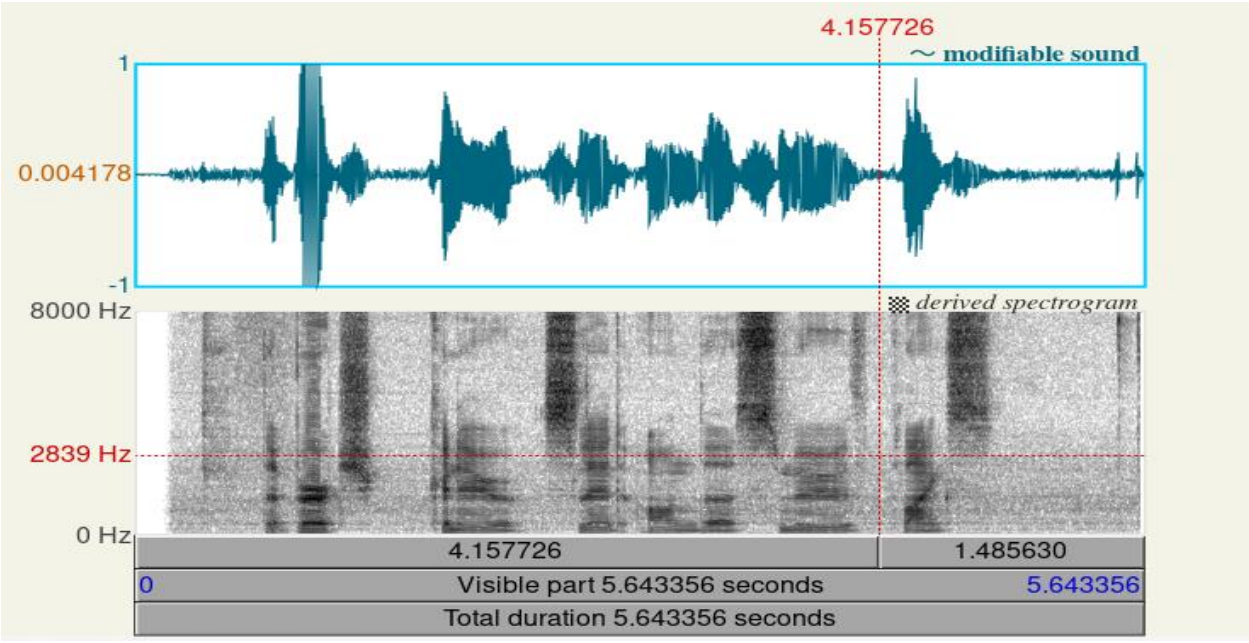


Figure 6. Narrowband spectrogram (window = 0.02 s). Horizontal harmonic striations are clearly resolved in voiced regions, confirming periodicity and allowing voicing to be identified from the spectrum.

4. How can you measure the fundamental frequency on the waveform, spectrogram and spectrum? Show one example and give the obtained values for the fundamental frequency and period.

F0 can be measured from all three representations as follows.

From the waveform: Zoom into a voiced segment until individual pitch cycles are visible. Select the time span between two adjacent equivalent peaks (one pitch period). In Figure 8, a selection of 0.009 s was made, giving T0 = 0.009 s and F0 = 1/0.009 = 111 Hz. This overestimates F0 compared to the pitch tracker value of 92.88 Hz (T0 = 10.77 ms). The real period is about 10.77 ms, so the 9 ms selection was around 1.77 ms too short. The most likely reason is that the two peaks I selected were not at exactly the same phase ,the waveform within each cycle is asymmetric, making it easy to pick slightly mismatched points. The waveform approach is only a rough estimate and the pitch tracker gives the more reliable result.

From the spectrogram: In the narrowband spectrogram, the frequency spacing between adjacent harmonic striations equals F0. Figure 7 shows the cursor positioned on the lowest clearly visible harmonic stripe, which reads 364 Hz on the left axis. This is the 4th harmonic (4 x 95 Hz = 380 Hz, approximately). The pitch tracker on the right confirms F0 = 95.33 Hz.

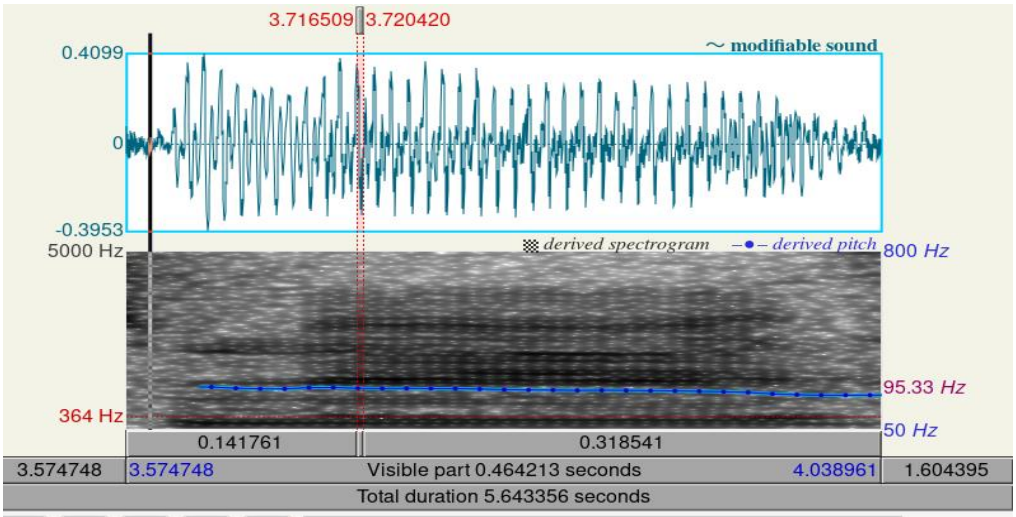


Figure 7. Cursor on the lowest visible harmonic stripe in the narrowband spectrogram of the vowel in "smooth". The stripe reads 364 Hz (4th harmonic). The pitch tracker on the right gives F0 = 95.33 Hz directly.

From the spectrum: F0 can also be read from the spectral slice as the spacing between adjacent harmonic peaks. In Figure 3, the peaks are spaced approximately 95 Hz apart, consistent with the pitch tracker reading of 95.33 Hz.

From the pitch tracker: Praat's built-in pitch tracker displays F0 as a blue curve. Clicking on the curve in the vowel region reads F0 = 95.33 Hz directly, giving T0 = 1/95.33 = 10.49 ms. This is the most accurate of the three methods. The spectrogram and pitch tracker both give approximately 95 Hz, while the waveform hand measurement gives 111 Hz — the small discrepancy is expected given the difficulty of selecting exact equivalent peaks manually.

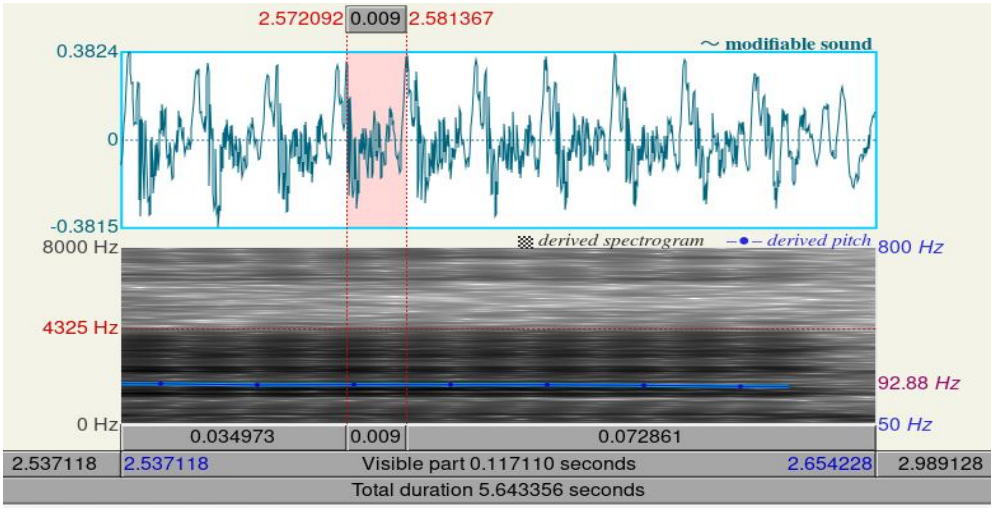


Figure 8. Period measurement in the zoomed waveform of /uː/ in "smooth". The manually selected interval of 0.009 s yields an estimated F0 of 111 Hz, a slight overestimate relative to the pitch tracker value of 92.88 Hz visible at right, attributable to imprecise peak selection by hand.

5. Using a long window can be disadvantageous for some specific tasks. In which cases can it be problematic? i.e., if you can choose between a window of length 3-4 periods or a window of 10-20 periods, when may it be of interest to use a shorter window?

A long window is problematic whenever the signal is changing quickly, because the window ends up averaging over sounds that are acoustically different. A shorter window makes more sense in those cases.

For example, at a consonant-vowel boundary like /k/ into the vowel in “canoe”, a 20 ms window spans both and mixes their spectra together. The same goes for stop bursts — the /p/ or /t/ release lasts only a few milliseconds, and a long window spreads it into the surrounding silence so it barely shows up. Diphthongs are another case where it matters: the formants move quickly enough that a long window just blurs the trajectory rather than tracking it. So basically, whenever timing is more important than frequency resolution, the shorter window is the better choice.

4 Formants

In this exercise, you will obtain the values of the first two formants of (at least) 5 vowels. You can use your own recordings. For example, you can record the vowels in your mother tongue. Be sure that you record stable vowels to make your measurements easier. If your selected language has more than five vowels, give at least the values for 5 of them.

Alternatively, you can download the English vowels from: https://linguistics.ucla.edu/people/hayes/103/Charts/VChart/#TheVowels

Note: own recording of Spanish vowels /a/, /e/, /i/, /o/, /u/ sustained individually, recorded at 44,100 Hz in Praat.

1. Select stable or central points inside each vowel and measure the first two formants and their bandwidths. Fill in Table 1. The BW should not be bigger than 25% of the formant value (BW/Formant < 0.25).

For each vowel, the cursor was placed at a stable central point where the formant tracks (red dots) were flat and the intensity was at its peak. Formant values and bandwidths were obtained using the Formant menu in Praat (Get first/second formant, Get bandwidth of first/second formant). The bandwidth quality check BW/Formant < 0.25 was applied to all measurements.

Table 1 -- Vowel Formants and Bandwidths

Vowel	F1 (Hz)	BW1 (Hz)	F2 (Hz)	BW2 (Hz)
/a/	759	91	1288	216
/e/	502	101	1787	292
/i/	305	26	1865	226
/o/	481	72	837	84
/u/	402	85	802	82

Table 1. Measured F1, BW1, F2 and BW2 for Spanish vowels /a e i o u/. All BW/Formant ratios are below 0.25.

Figures 1 to 5 show the Praat spectrogram for each vowel with the cursor positioned at the stable measurement point and the formant tracks (red dots) visible. Figure 7 shows an example of the Praat Info window output for vowel /a/, displaying the mean F1 value of 776.91 Hz returned by the Get first formant command. All five vowels were measured in the same way.

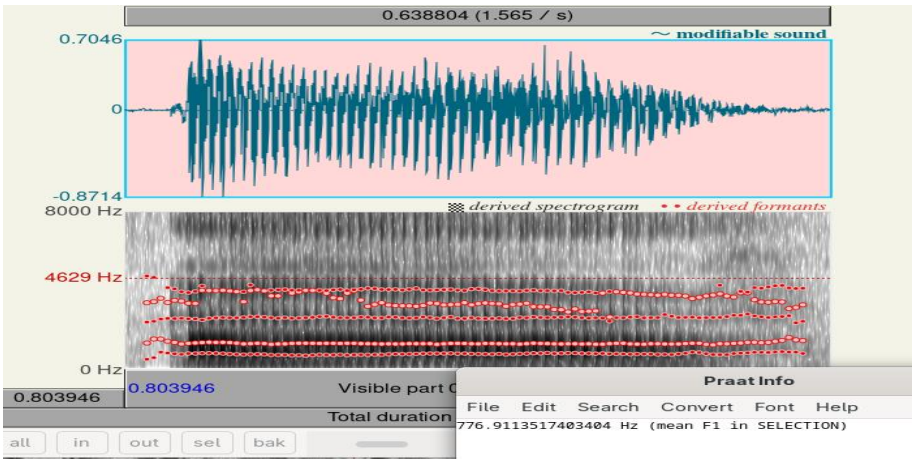


Figure 7. Praat Info window showing the mean F1 output for vowel /a/ (776.91 Hz). The value was obtained using Formant → Get first formant with the cursor placed at the stable centre of the vowel.

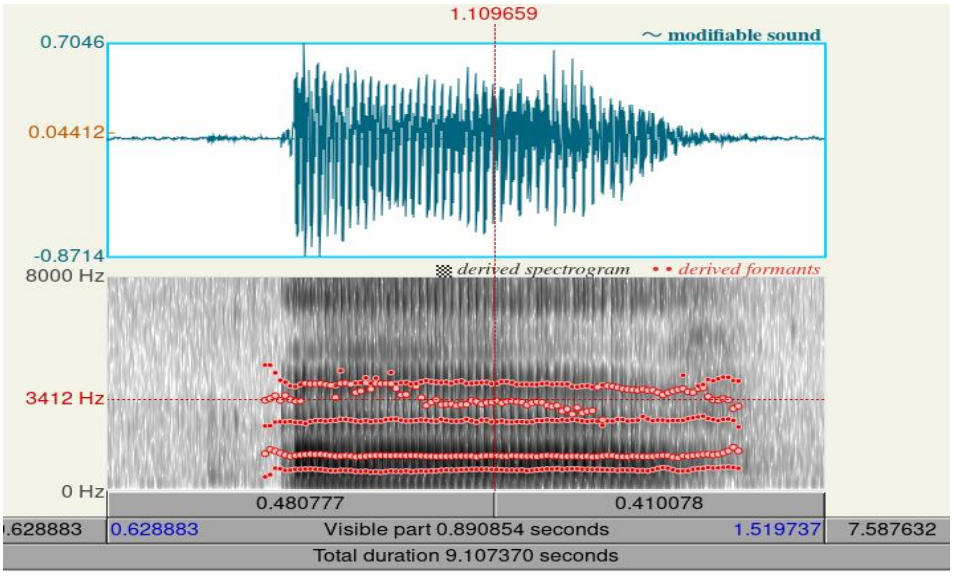


Figure 1. Waveform (top) and narrowband spectrogram (bottom) of the vowel /a/, with red formant tracks overlaid. Cursor placed at the stable centre of the vowel, reading F1 ≈ 759 Hz and F2 ≈ 1288 Hz.

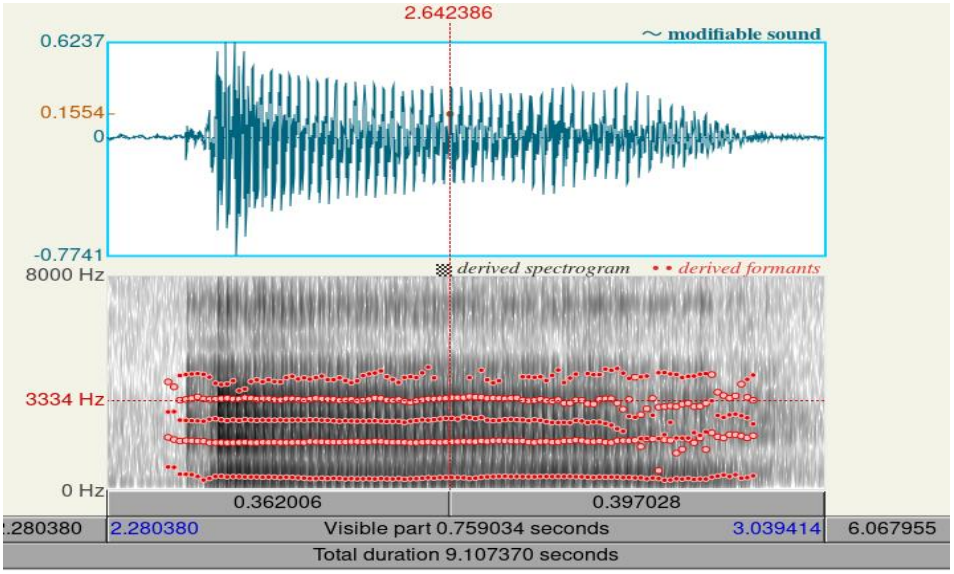


Figure 2. Waveform (top) and narrowband spectrogram (bottom) of the vowel /e/, with red formant tracks overlaid. Cursor placed at the stable centre of the vowel, reading F1 ≈ 502 Hz and F2 ≈ 1787 Hz.

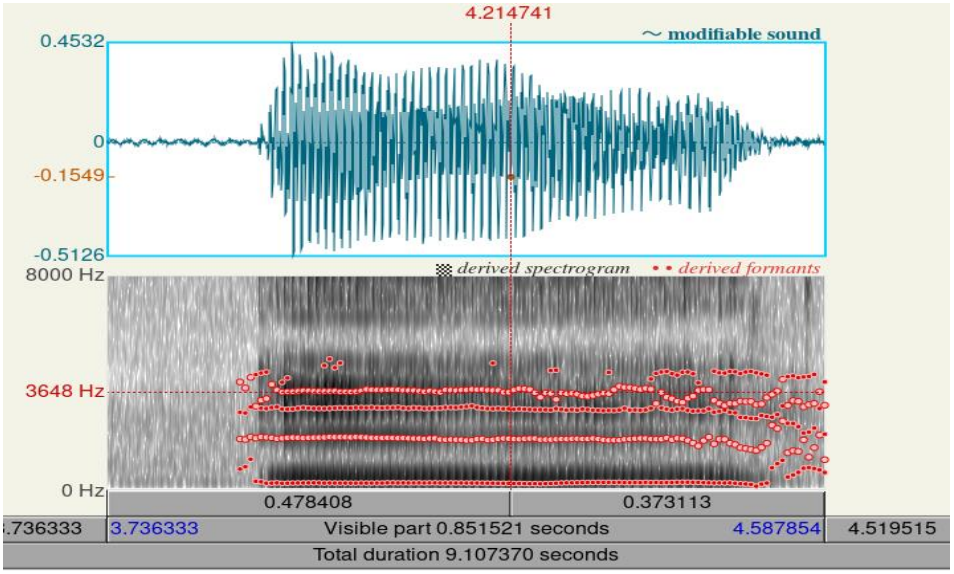


Figure 3. Waveform (top) and narrowband spectrogram (bottom) of the vowel /i/, with red formant tracks overlaid. Cursor placed at the stable centre of the vowel, reading F1 ≈ 305 Hz and F2 ≈ 1865 Hz.

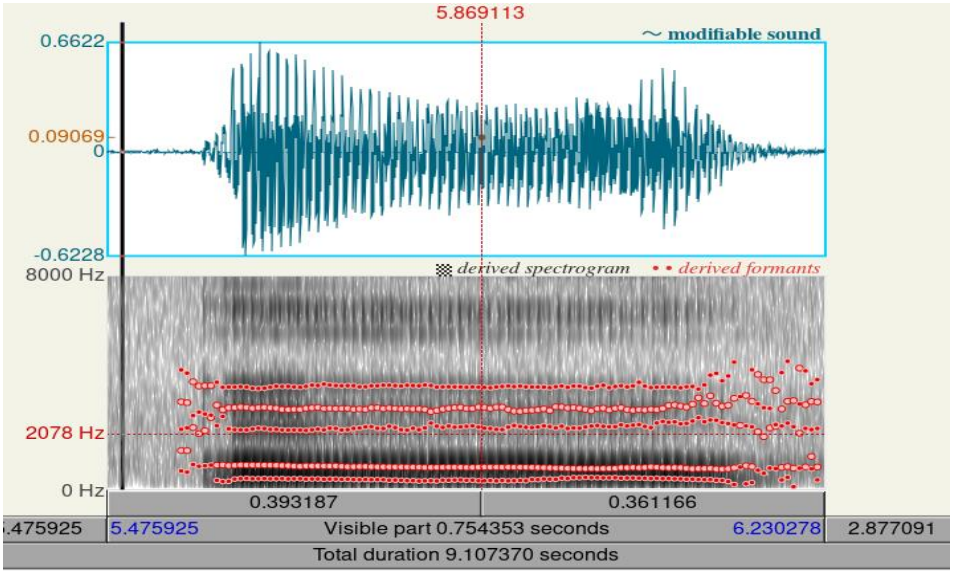


Figure 4. Waveform (top) and narrowband spectrogram (bottom) of the vowel /o/, with red formant tracks overlaid. Cursor placed at the stable centre of the vowel, reading F1 ≈ 481 Hz and F2 ≈ 837 Hz.

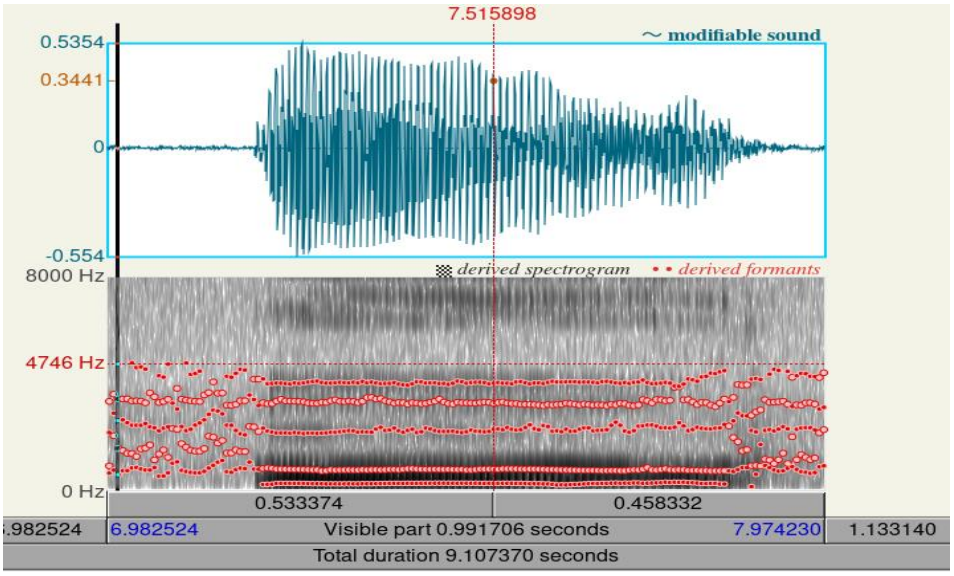


Figure 5. Waveform (top) and narrowband spectrogram (bottom) of the vowel /u/, with red formant tracks overlaid. Cursor placed at the stable centre of the vowel, reading F1 ≈ 402 Hz and F2 ≈ 802 Hz.

2. Represent the values of the formants in a graphic of your choice (F1 vs F2, vowel vs frequency for F1 & F2...).

Figure 6 shows the F1 vs F2 vowel space for the five Spanish vowels. Both axes are inverted following phonetic convention, so the plot roughly mirrors how the vowels sit in the mouth — high vowels at the top, back vowels to the left.

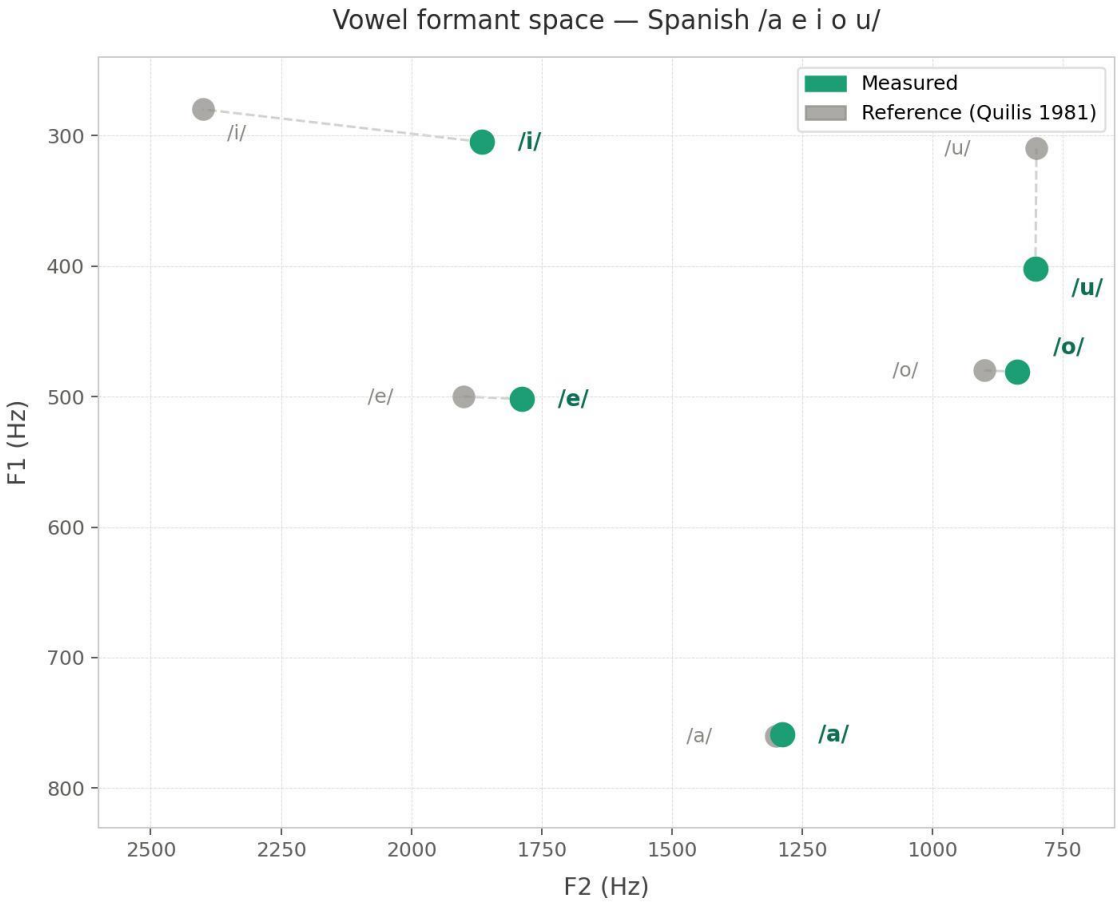


Figure 6. F1 vs F2 vowel space for Spanish /a e i o u/. Green dots are measured values; grey dots are reference values from Quilis (1981). Both axes are inverted following standard phonetic convention.

3. Compare your obtained values with standard values (research the standard values of the formants in the chosen language). Do they agree?

The measured values were compared against standard Spanish formant values from Quilis (1981), which gives the following reference ranges for male speakers: /a/ F1 760 Hz, F2 1300 Hz; /e/ F1 500 Hz, F2 1900 Hz; /i/ F1 280 Hz, F2 2400 Hz; /o/ F1 480 Hz, F2 900 Hz; /u/ F1 310 Hz, F2 800 Hz.

Overall the results agree fairly well with the Quilis values. The F1 ranking is exactly right (/a/ highest, /i/ lowest) which matches the expected pattern where F1 increases as the tongue lowers. The F2 ranking is also correct, with the front vowels /i/ and /e/ having higher F2 than the back vowels /o/ and /u/.

The clearest mismatch is /i/ F2, which came out at 1865 Hz instead of ~2400 Hz. This is probably a mix of speaker variation (which can easily account for a few hundred Hz)

and a recording effect, since sustained isolated vowels tend to produce lower F2 for /i/ than in running speech. The /u/ F1 is also a bit high at 402 Hz versus the reference of 310 Hz. Part of this is that Quilis gives values for male speakers, and as a female speaker my formants are generally shifted upward. The sustained production may also have caused a slightly more open articulation than usual, which would raise F1. Overall the values are close enough to the reference that the measurements look reasonable.

5 Reading spectrograms

The spectrogram in the next page corresponds to a speech utterance. Analyze it and answer to the following questions, explaining your responses.

1. What is the sampling frequency of the signal? How have you found it?

The sampling frequency of the signal is 16000 Hz. The frequency axis reaches up to 8 kHz, which corresponds to the Nyquist frequency. Therefore, fs is 8000 x 2 = 16000 Hz.

2. Is it a narrow-band or a wide-band spectrogram?

It is a narrowband spectrogram. This is evident from the clearly resolved horizontal harmonic striations visible throughout the voiced regions. These striations can only be distinguished when the analysis window is long enough to provide sufficient frequency resolution to separate individual harmonics. A wideband spectrogram, with its shorter window, would instead show vertical striations corresponding to individual glottal pulses, without resolving the individual harmonics.

3. Is it a male or a female voice? What is its approximate pitch?

It is a female voice with an approximate pitch of 200 Hz. This was estimated by measuring the vertical spacing between adjacent harmonic striations in the narrowband spectrogram, which directly corresponds to F0. The typical range for a female speaking voice is 165–255 Hz, and the spacing of the horizontal striations is consistent with approximately 200 Hz. The F0 value also places this speaker above the typical male range (85–180 Hz), confirming a female voice.

4. Look at the spectrogram in the fragment that goes from 0.74 to 0.80 seconds. Does it correspond to one or more than one phones? Is it/are they voiced or unvoiced? Can you roughly say where the formants are?

The fragment appears to correspond to a single phone, since its structure is uniform without abrupt changes. It is a voiced sound, as clear horizontal parallel lines are visible. The formants are approximately located at: F1 in 400 Hz, F2 in 1000 Hz, and F3 in 2000 Hz.

5. Can you distinguish an unvoiced phone in the signal? Provide the instant of time at which it begins and ends. Can you guess which type of consonant it is?

Yes, an unvoiced phone can be identified approximately between 0.85 seconds and 0.95 seconds. In this region, the harmonic structure characteristic of voiced sounds clearly disappears. These features (absence of harmonics, low but non zero energy, and a duration of approximately 100 ms) suggest that this phone is most likely an unvoiced fricative.

6. Is this signal recorded in a noisy background? Which is the approximate average power spectral density in the silent parts of the recording? And in the speech part?

Yes, the signal appears to have been recorded in a slightly noisy environment. In the silent parts of the recording (such as before speech starts and after it ends), the spectrogram does not show a completely dark background but rather a medium blue tone with some lighter diffuse patches, indicating that a low level of background noise is present. The approximate average power spectral density values are:

-Silent parts: approximately -90 to -100 db/Hz.

-Speech parts: approximately -30 to -40 db/Hz (in the most intense regions)

7. There is a darker band in the upper part of the spectrogram. Can you explain why this could happen?

This can be explained by the fact that the human voice has very little energy in the high-frequency range: most of the speech information is concentrated below 4kHz, with some additional contribution from unvoiced fricatives that can reach up to 8 kHz. Above these frequencies, the voice barely contributes any energy.